Stat 6750 Applied Project: Find the Optimum

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June 30, 2026

1 Experimental Strategy

1.1 Problem and constraints

The goal of this project is to maximize the yield (in percent) of a chemical process. The process has eight controllable additives, which I treat as factors

$$A, B, C, D, E, F, G, H.$$

Each factor can be set between 0 and 10 (natural units). The current operating condition used in practice is

$$A = B = C = D = 8, \qquad E = F = G = H = 0,$$

and this combination gives a yield of about 50%. Very little experimentation has been done away from this point, especially for factors E–H, which are currently not used. I have a total budget of 75 runs, and each submission to the instructor returns the true process yield for the proposed design.

All runs must keep the factor levels inside $[0, 10]$. I also treat each submission as one "day" of work; conditions are assumed stable within a submission but may change slightly from day to day.

1.2 Overall strategy for the 75 runs

My overall plan for the 75 runs is as follows.

1. **Initial screening (Submission 1).** Start with a regular 2^{8-4} fractional factorial design in all eight factors. This design is resolution IV, so main effects are clear of other main effects and two–factor interactions. The goal is to identify which factors have the strongest influence on yield and to locate a promising region of the factor space. This used 16 runs.

2. **Follow–up search in the important factors.** After screening, I focus on the factors that appear active. Based on Submission 1, these are the additives B, D, and F. I then run smaller 2^k designs in these factors, keeping the others fixed at a safe middle value. The goal is to move in a "steepest ascent" direction (for maximizing yield) while staying inside the $[0, 10]$ bounds and to push clearly harmful factors (such as D and F) to their best levels.

3. **Local modelling around the best region.** Once I am near a region with high yield, I switch to response-surface style designs in the most important factors. In this project I build a small face–centered central composite design (CCD) in three factors (B, E, G) around a high–yield center point, combine these runs with the earlier 2^3 design in (B, E, G) (in coded form), and fit a second–order model with main effects, interactions, and simple quadratic terms.

4. **Confirmation runs and prediction interval.** Near the end of the run budget I plan a small set of confirmation runs at the final recommended settings. These replicates provide a check on the fitted model and allow me to build a 95% prediction interval for the yield at the chosen operating point.

1.3 Summary of submissions

Over the course of the project I have completed six submissions. In total I used 58 runs out of the 75–run budget.

- **Submission 1:** a 16–run 2^{8-4} screening design in all eight factors. This identified B as having a strong positive effect and D and F as having strong negative effects, while the other factors appeared inactive in that region.

- **Submission 2:** a full 2^3 design in B, D, and F around the middle of the $[0, 10]$ range, with the other factors fixed at $A = C = E = G = H = 5$. This showed that yield is highest when $D = 0$ and $F = 0$ and when B is at its lower level within that cube ($B = 7.5$ instead of $B = 10$).

- **Submission 3:** a small follow-up around the promising setting with $D = 0$ and $F = 0$, comparing $B = 5$ and $B = 7.5$ with replication. This was used to refine the level of B and to confirm that we are already in a high-yield region with yields close to 90%.

- **Submission 4:** a 2^3 design in (B, E, G) with $B \in \{4, 6\}$, $E \in \{2.5, 7.5\}$, $G \in \{2.5, 7.5\}$ and four center runs at $B = E = G = 5$, keeping $A = C = H = 5$ and $D = F = 0$. This explored how E and G behave near the good setting found in Submissions 2 and 3 and started building a face–centered CCD in (B, E, G). The highest yield in this submission (about 92%) was observed at $B = 4$, $E = 2.5$, $G = 2.5$.

- **Submission 5:** six axial points in (B, E, G) at $(\pm1, 0, 0)$, $(0, \pm1, 0)$, and $(0, 0, \pm1)$ in coded units around the center $(B, E, G) = (5, 5, 5)$, with $A = C = H = 5$ and $D = F = 0$. Combined with Submission 4, these runs complete a small face–centered CCD in (B, E, G) and allow a second–order model to be fitted in this local high–yield region. The best yield in the CCD (92.67%) occurs at $B = 5$, $E = 7.5$, $G = 5$.

- **Submission 6:** 12 replicate runs at the high–yield setting $A = C = H = 5$, $D = F = 0$, $B = 5$, $E = 7.5$, $G = 5$. These replicates are used to check stability at the chosen operating point and to build a 95% confidence interval for the mean yield and a 95% prediction interval for a single future run.

After Submission 6 I still had some runs left in the 75–run budget. I decided not to use the remaining runs because the CCD and replicate runs already showed a broad flat high–yield region and a stable prediction interval; additional runs would mainly give more precision, rather than change the conclusion.

Section 3.1 gives the detailed analysis for Submission 1, and the later subsections in Section 3 give the details for Submissions 2–6.

2 Final model and recommended settings

Using the 18 runs from Submissions 4 and 5 (a small face–centered CCD in B, E, G with $A = C = H = 5$ and $D = F = 0$), I fitted the following quadratic response–surface model in coded variables x_B, x_E, x_G. The coding is

$$B : 4 \mapsto -1,\ 5 \mapsto 0,\ 6 \mapsto +1, \qquad E, G : 2.5 \mapsto -1,\ 5 \mapsto 0,\ 7.5 \mapsto +1.$$

The fitted model (predicted yield $\hat{y}$ in percent) is

$$\begin{aligned} \hat{y} &= 90.66 - 0.80\, x_B - 0.26\, x_E + 0.62\, x_G \\ &\quad - 1.26\, x_B^2 + 1.13\, x_E^2 - 0.84\, x_G^2 \\ &\quad + 0.45\, x_B x_E + 0.54\, x_B x_G + 0.24\, x_E x_G. \end{aligned}$$

Contour and surface plots of this model show a fairly flat high–yield ridge in (B, E, G) rather than a sharp single optimum. Along this ridge, the simple setting

$$B = 5, \quad E = 7.5, \quad G = 5$$

has a high predicted mean yield (about 91.5% from the quadratic model) and agrees with the best observed run in the CCD.

Combining this with the earlier screening results, my final recommended operating settings for all eight factors are

$$(A, B, C, D, E, F, G, H) = (5, 5, 5, 0, 7.5, 0, 5, 5).$$

To check this choice, Submission 6 consisted of 12 replicate runs at this setting. The sample mean and sample standard deviation of the 12 yields were

$$\bar{y}_6 = 90.88\%, \qquad s_6 = 1.47\%.$$

Using these replicates, a 95% prediction interval for the yield of a single future run at the recommended setting is

$$87.51\% \ \leq\ Y_{\text{new}}\ \leq\ 94.25\%.$$

Thus the process is expected to give a stable yield of about 91%, with most individual runs falling inside this range.

3 Detailed analyses by submission

In this section I show the key R outputs and plots that support the interpretations in Section 1 and the final conclusions in Section 2. For each submission (1–6) I include the most important pieces of output that were used to plan the next runs. At the end of this section I also summarize the analysis of the final quadratic model in (B, E, G) that was used to choose the recommended settings. Only the parts that are directly relevant to the conclusions are included.

3.1 Submission 1: 2^{8-4} screening design

Design and response

Table 1 shows the design matrix for Submission 1 together with the observed yield `ali1`. Each factor was run at 2.5 (coded -1) and 7.5 (coded $+1$).

Table 1: Submission 1: design matrix and response (raw R output)

```
> cbind(A,B,C,D,E,F,G,H,ali1)
        A   B   C   D   E   F   G   H  ali1
 [1,] 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 50.34
 [2,] 2.5 2.5 2.5 7.5 2.5 7.5 7.5 7.5 30.35
 [3,] 2.5 2.5 7.5 2.5 7.5 2.5 7.5 7.5 49.34
 [4,] 2.5 2.5 7.5 7.5 7.5 7.5 2.5 2.5 26.21
 [5,] 2.5 7.5 2.5 2.5 7.5 7.5 2.5 7.5 53.75
 [6,] 2.5 7.5 2.5 7.5 7.5 2.5 7.5 2.5 46.97
 [7,] 2.5 7.5 7.5 2.5 2.5 7.5 7.5 2.5 55.09
 [8,] 2.5 7.5 7.5 7.5 2.5 2.5 2.5 7.5 47.70
 [9,] 7.5 2.5 2.5 2.5 7.5 7.5 7.5 2.5 34.24
[10,] 7.5 2.5 2.5 7.5 7.5 2.5 2.5 7.5 41.55
[11,] 7.5 2.5 7.5 2.5 2.5 7.5 2.5 7.5 33.90
[12,] 7.5 2.5 7.5 7.5 2.5 2.5 7.5 2.5 42.27
[13,] 7.5 7.5 2.5 2.5 2.5 2.5 7.5 7.5 52.14
[14,] 7.5 7.5 2.5 7.5 2.5 7.5 2.5 2.5 44.15
[15,] 7.5 7.5 7.5 2.5 7.5 2.5 2.5 2.5 52.99
[16,] 7.5 7.5 7.5 7.5 7.5 7.5 7.5 7.5 45.48
```

Main-effect estimates

The main-effect estimates from the 2^{8-4} design (R output from code #2.1) are:

```
> effects_main
       A        B        C        D        E        F        G        H
-1.62875 11.25875 -0.06375 -7.13875 -0.67625 -7.51625  0.66125  0.24375
```

These values are used in Section 1 to argue that B has a large positive effect and that D and F have large negative effects.

Half-normal plot of effects

Figure 1 shows the half-normal plot of all main and two-factor effects. Most effects lie near the line, while B, F, D and the alias group $AD = BF = CG = EH$ are far from the line.

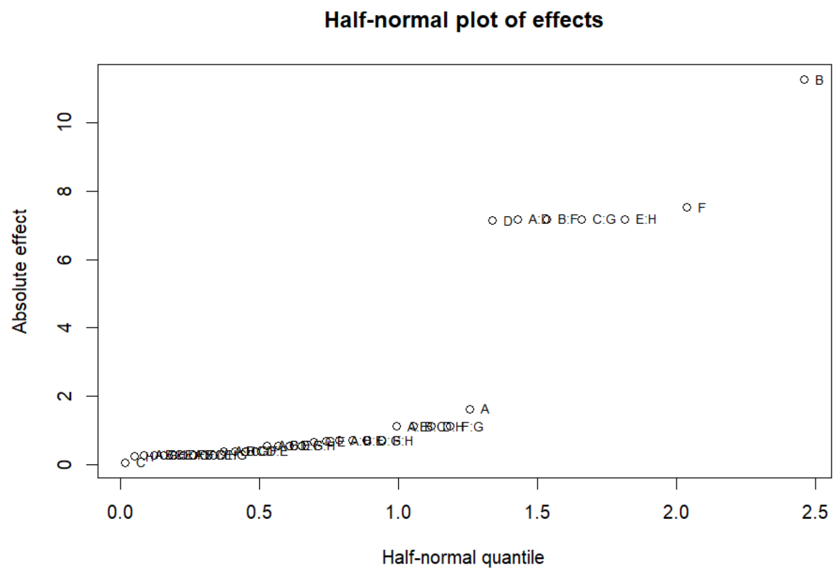

Figure 1: Submission 1: half-normal plot of effects (R output from code #3.1).

Lenth's method output

For the unreplicated design, Lenth's method was used to identify active effects. The PSE and margins were:

```
> s0; PSE; ME; SME
[1] 0.915
[1] 0.838125
[1] 2.095312
[1] 2.933437
```

The sorted Lenth table (R output from code #4.1) is:

```
> Lenth_table[order(-Lenth_table$abs_est), ]
    effect estimate  abs_est     t_Lenth sig_ME sig_SME
B        B 11.25875 11.25875 13.43325876   TRUE    TRUE
F        F -7.51625  7.51625 -8.96793438   TRUE    TRUE
A:D    A:D  7.18375  7.18375  8.57121551   TRUE    TRUE
B:F    B:F  7.18375  7.18375  8.57121551   TRUE    TRUE
C:G    C:G  7.18375  7.18375  8.57121551   TRUE    TRUE
E:H    E:H  7.18375  7.18375  8.57121551   TRUE    TRUE
D        D -7.13875  7.13875 -8.51752424   TRUE    TRUE
A        A -1.62875  1.62875 -1.94332588  FALSE   FALSE
A:E    A:E  1.12625  1.12625  1.34377330  FALSE   FALSE
B:C    B:C  1.12625  1.12625  1.34377330  FALSE   FALSE
D:H    D:H  1.12625  1.12625  1.34377330  FALSE   FALSE
F:G    F:G  1.12625  1.12625  1.34377330  FALSE   FALSE
A:C    A:C  0.70375  0.70375  0.83967189  FALSE   FALSE
B:E    B:E  0.70375  0.70375  0.83967189  FALSE   FALSE
```

```
D:G    D:G  0.70375  0.70375  0.83967189  FALSE   FALSE
F:H    F:H  0.70375  0.70375  0.83967189  FALSE   FALSE
E        E -0.67625  0.67625 -0.80686055  FALSE   FALSE
G        G  0.66125  0.66125  0.78896346  FALSE   FALSE
A:B    A:B -0.55875  0.55875 -0.66666667  FALSE   FALSE
C:E    C:E -0.55875  0.55875 -0.66666667  FALSE   FALSE
D:F    D:F -0.55875  0.55875 -0.66666667  FALSE   FALSE
G:H    G:H -0.55875  0.55875 -0.66666667  FALSE   FALSE
A:H    A:H -0.38875  0.38875 -0.46383296  FALSE   FALSE
B:G    B:G -0.38875  0.38875 -0.46383296  FALSE   FALSE
C:F    C:F -0.38875  0.38875 -0.46383296  FALSE   FALSE
D:E    D:E -0.38875  0.38875 -0.46383296  FALSE   FALSE
A:F    A:F -0.27875  0.27875 -0.33258762  FALSE   FALSE
B:D    B:D -0.27875  0.27875 -0.33258762  FALSE   FALSE
C:H    C:H -0.27875  0.27875 -0.33258762  FALSE   FALSE
E:G    E:G -0.27875  0.27875 -0.33258762  FALSE   FALSE
A:G    A:G -0.27625  0.27625 -0.32960477  FALSE   FALSE
B:H    B:H -0.27625  0.27625 -0.32960477  FALSE   FALSE
C:D    C:D -0.27625  0.27625 -0.32960477  FALSE   FALSE
E:F    E:F -0.27625  0.27625 -0.32960477  FALSE   FALSE
H        H  0.24375  0.24375  0.29082774  FALSE   FALSE
C        C -0.06375  0.06375 -0.07606264  FALSE   FALSE
```

These outputs support the conclusion that B, D, and F and the alias group $AD = BF = CG = EH$ are active, while other effects are negligible.

First-order model output

The first-order model in coded variables (R output from `summary(fit_fo)`) is:

```
> summary(fit_fo)

Call:
lm(formula = y ~ xA + xB + xC + xD + xE + xF + xG + xH, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-4.9794 -3.3181  0.0531  3.7806  4.1656

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 44.15437    1.38857  31.799 7.86e-09 ***
xA          -0.81437    1.38857  -0.586  0.57597
xB           5.62938    1.38857   4.054  0.00485 **
xC          -0.03187    1.38857  -0.023  0.98233
xD          -3.56938    1.38857  -2.571  0.03698 *
xE          -0.33812    1.38857  -0.244  0.81460
xF          -3.75813    1.38857  -2.706  0.03035 *
xG           0.33063    1.38857   0.238  0.81862
```

```
xH           0.12188    1.38857   0.088  0.93252
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.554 on 7 degrees of freedom
Multiple R-squared:  0.815, Adjusted R-squared:  0.6036
F-statistic: 3.855 on 8 and 7 DF,  p-value: 0.04603
```

The ANOVA table for this model (R output from `anova(fit_fo)`) is:

```
> anova(fit_fo)
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value   Pr(>F)
xA         1  10.61   10.61  0.3440 0.575965
xB         1 507.04  507.04 16.4356 0.004846 **
xC         1   0.02    0.02  0.0005 0.982327
xD         1 203.85  203.85  6.6077 0.036978 *
xE         1   1.83    1.83  0.0593 0.814595
xF         1 225.98  225.98  7.3250 0.030353 *
xG         1   1.75    1.75  0.0567 0.818619
xH         1   0.24    0.24  0.0077 0.932517
Residuals  7 215.95   30.85
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

These outputs are the basis for the discussion in Section 1 that B, D, and F are the important factors in this region and that the model explains about 81.5% of the variation.

Residual plots

Finally, Figure 2 shows the residuals versus fitted values and the normal Q–Q plot for the first-order model.

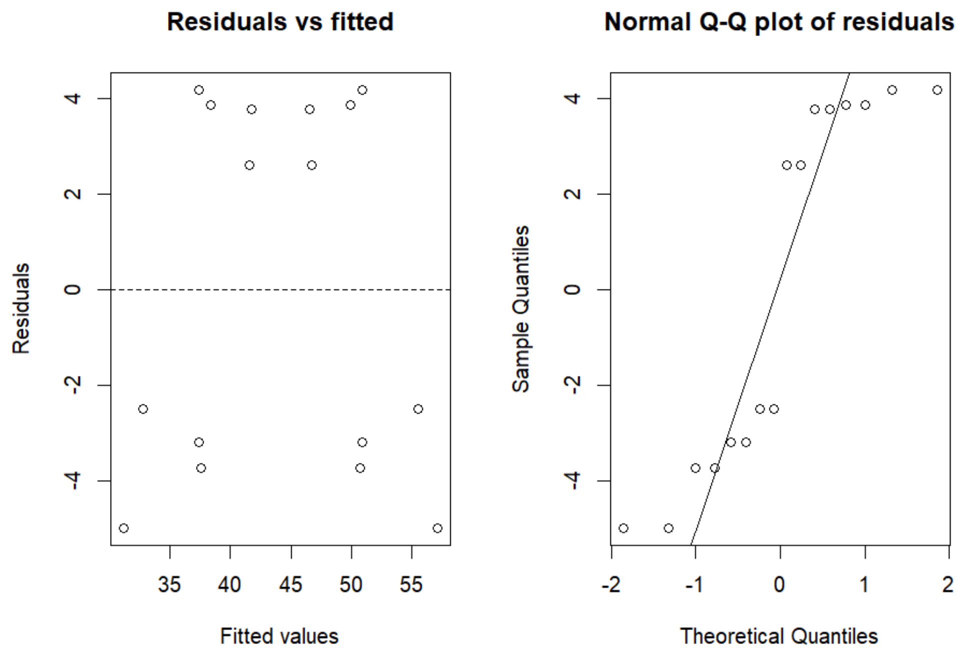

Figure 2: Submission 1: residuals vs fitted and normal Q–Q plot (R output from code #5.3).

These plots support the conclusion that the first-order model is adequate as a directional model for planning the next submission.

3.2 Submission 2: 2^3 follow-up in B, D, and F

After Submission 1 I focused on the three additives that looked most important: B (strong positive effect), D (negative effect), and F (negative effect). In Submission 2 I ran a full 2^3 factorial in these three factors, keeping all other factors fixed at a middle value:

$$A = C = E = G = H = 5.$$

I chose the ranges

$$B \in \{7.5, 10\}, \quad D \in \{0, 2.5\}, \quad F \in \{0, 2.5\},$$

to explore a region around the middle of the $[0, 10]$ interval while staying inside the bounds.

Design and response

Table 2 shows the design matrix and the observed yield `ali2`. Only B, D, and F change in this design; the other five factors are held at $A = C = E = G = H = 5$.

Table 2: Submission 2: design matrix and response (raw R output)

```
> cbind(A2,B2,C2,D2,E2,F2,G2,H2,y2)
     A2   B2 C2  D2 E2  F2 G2 H2    y2
[1,]  5  7.5  5 0.0  5 0.0  5  5 86.29
[2,]  5  7.5  5 0.0  5 2.5  5  5 77.01
[3,]  5  7.5  5 2.5  5 0.0  5  5 60.38
[4,]  5  7.5  5 2.5  5 2.5  5  5 52.97
[5,]  5 10.0  5 0.0  5 0.0  5  5 73.26
[6,]  5 10.0  5 0.0  5 2.5  5  5 68.84
[7,]  5 10.0  5 2.5  5 0.0  5  5 50.37
[8,]  5 10.0  5 2.5  5 2.5  5  5 45.05
```

The yields range from about 45% to 86%. In particular, the best run (86.29%) is a large improvement over the original baseline of about 50%.

Effect estimates

Using coded variables $x_B, x_D, x_F \in \{-1, +1\}$ for the low and high levels, the main-effect and interaction estimates (R output from code #S2.2) are:

```
> effects_main2
       B        D        F
 -9.7825 -24.1575  -6.6075

> effects_2f2
    BD     BF     DF
0.8175 1.7375 0.2425

> effect_3f2
[1] -0.6925

> effects_all2
       B        D        F       BD       BF       DF      BDF
 -9.7825 -24.1575  -6.6075   0.8175   1.7375   0.2425  -0.6925
```

The negative signs of the main effects mean that, in this region, increasing B, D, or F from their low level to their high level decreases the yield. The effect of D is especially large (about -24 percentage points), followed by B and then F. All interaction effects are small compared to the main effects.

Half-normal plot of effects

Figure 3 shows the half-normal plot of the seven effects (B, D, F, BD, BF, DF, and BDF). The points for D, B, and F lie far from the reference line, while the interaction effects lie close to the line.

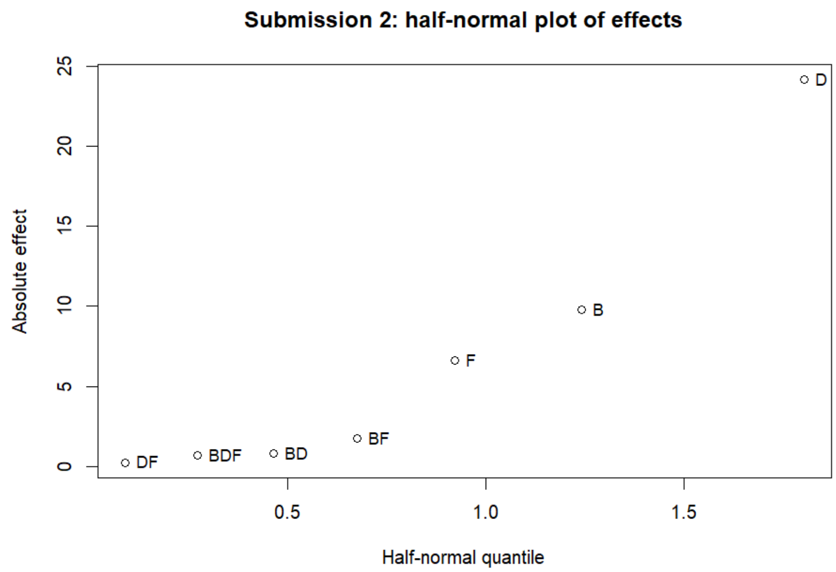

Figure 3: Submission 2: half-normal plot of effects (R output from code #S2.3).

This visual impression agrees with the numerical effects: D, B, and F are active, and there is no strong evidence of important interactions in this region.

Lenth's method output

For completeness I also applied Lenth's method to this unreplicated 2^3 design. The PSE and margins (R output from code #S2.4) are:

```
> s0_2; PSE_2; ME_2; SME_2
[1] 2.60625
[1] 1.1325
[1] 2.83125
[1] 3.476775
```

The Lenth table, sorted by absolute effect size, is:

```
> Lenth_table2[order(-Lenth_table2$abs_est), ]
    effect estimate abs_est     t_Lenth sig_ME sig_SME
D        D -24.1575 24.1575 -21.3311258   TRUE    TRUE
B        B  -9.7825  9.7825  -8.6379691   TRUE    TRUE
F        F  -6.6075  6.6075  -5.8344371   TRUE    TRUE
BF      BF   1.7375  1.7375   1.5342163  FALSE   FALSE
BD      BD   0.8175  0.8175   0.7218543  FALSE   FALSE
BDF    BDF  -0.6925  0.6925  -0.6114790  FALSE   FALSE
DF      DF   0.2425  0.2425   0.2141280  FALSE   FALSE
```

All three main effects (D, B, and F) are flagged as significant by both the ME and SME cutoffs. None of the interaction terms are flagged. This supports the simple interpretation that only the main effects are important in this local region.

First-order model output

A first-order model with only the main effects in coded form (R output from `summary(fit2_fo)` and `anova(fit2_fo)`) is:

```
> summary(fit2_fo)

Call:
lm(formula = y2 ~ xB2 + xD2 + xF2, data = dat2)

Residuals:
      1       2       3       4       5       6       7       8
 1.7450 -0.9275 -0.0075 -0.8100 -1.5025  0.6850 -0.2350  1.0525

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  64.2712     0.5139 125.064 2.45e-08 ***
xB2          -4.8912     0.5139  -9.518  0.00068 ***
xD2         -12.0788     0.5139 -23.504 1.94e-05 ***
xF2          -3.3038     0.5139  -6.429  0.00301 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.454 on 4 degrees of freedom
Multiple R-squared:  0.9942, Adjusted R-squared:  0.9898
F-statistic: 228.1 on 3 and 4 DF,  p-value: 6.319e-05

> anova(fit2_fo)
Analysis of Variance Table

Response: y2
          Df  Sum Sq Mean Sq F value    Pr(>F)
xB2        1  191.39  191.39  90.589 0.0006803 ***
xD2        1 1167.17 1167.17 552.431 1.943e-05 ***
xF2        1   87.32   87.32  41.328 0.0030106 **
Residuals  4    8.45    2.11
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

All three main effects are highly significant, and the model explains 99.4% of the variation in the eight responses. The estimated regression function in coded variables is

$$\hat{y} = 64.27 - 4.89\,x_B - 12.08\,x_D - 3.30\,x_F.$$

Because the coefficients for x_B, x_D, and x_F are all negative, this model says that the yield increases as we move to lower levels of B, D, and F in this region.

Residual plots

Figure 4 shows the residuals versus fitted values and the normal Q–Q plot for the first-order model.

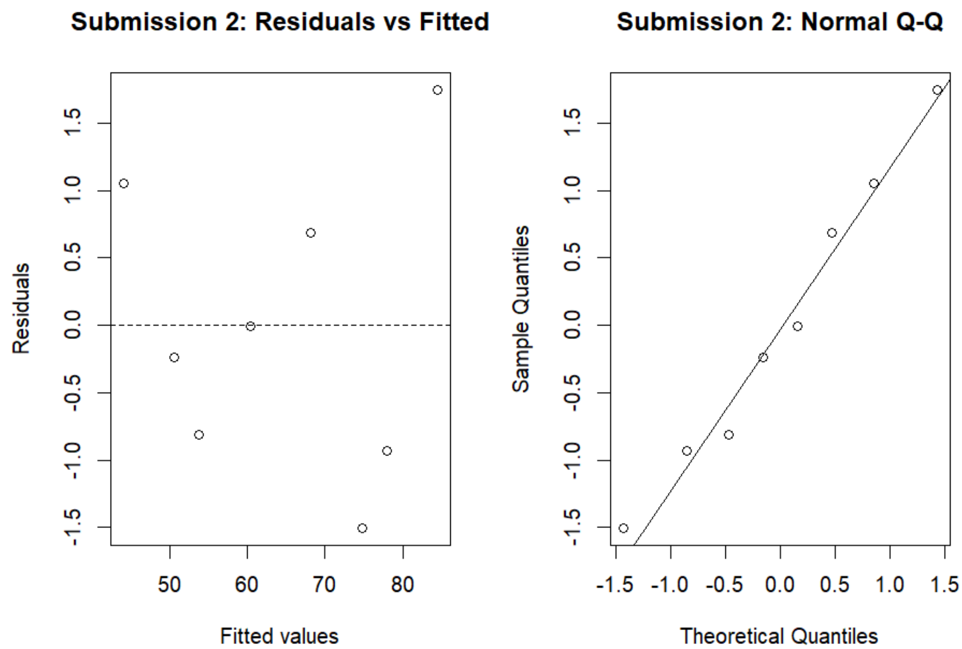

Figure 4: Submission 2: residuals vs fitted and normal Q–Q plot (R output from code #S2.6).

The residuals are small and show no clear pattern, and the Q–Q plot is reasonably close to a straight line. This suggests that the first-order model with only the main effects of B, D, and F is adequate in this local region.

Summary of Submission 2. Submission 2 shows that, around $A = C = E = G = H = 5$, the yield is highest when $D = 0$ and $F = 0$ (their lower bounds), and when B is at its lower level within this cube ($B = 7.5$ rather than $B = 10$). The best observed combination in this design is

$$B = 7.5, \quad D = 0, \quad F = 0,$$

with yield 86.29%. Combining this with the positive effect of B from Submission 1 suggests curvature in B: the response increases as B moves up from 2.5 to around 7–8, and then decreases when B is pushed toward 10. This reasoning motivated Submission 3, which focuses on refining the level of B while fixing $D = 0$ and $F = 0$.

3.3 Submission 3: refining B with replication at $D = F = 0$

The results from Submission 2 suggested that, around $A = C = E = G = H = 5$, the yield is highest when $D = 0$ and $F = 0$ and when B is closer to 7.5 than to 10. However, Submission 1 had shown that B was beneficial when moving up from the original low level (2.5). Together, this suggested possible curvature in B: the yield increases as B moves up from 2.5, but then drops again as B is pushed too high.

In Submission 3 I therefore fixed

$$A = C = E = G = H = 5, \qquad D = 0, \qquad F = 0,$$

and compared two nearby levels of B,

$$B \in \{5, 7.5\},$$

with two replicates at each level. The goals were (i) to check whether moving from $B = 7.5$ down to $B = 5$ gives any further improvement, and (ii) to obtain a small pure-error estimate in this high-yield region.

Design and response

Table 3 shows the design matrix and the observed yield `ali3`. Only B changes; all other factors are fixed at their chosen values.

Table 3: Submission 3: design matrix and response (raw R output)

```
> cbind(A3,B3,C3,D3,E3,F3,G3,H3,y3)
     A3  B3 C3 D3 E3 F3 G3 H3    y3
[1,]  5 5.0  5  0  5  0  5  5 91.61
[2,]  5 7.5  5  0  5  0  5  5 85.51
[3,]  5 5.0  5  0  5  0  5  5 88.30
[4,]  5 7.5  5  0  5  0  5  5 86.68
```

Using the coded variable $x_B \in \{-1, +1\}$ for $B = 5$ and $B = 7.5$, the mean responses at the two levels (R output from code #S3.2) are:

```
> tapply(y3, B3, mean)
     5    7.5
89.955 86.095

> tapply(y3, xB3, mean)
    -1      1
89.955 86.095

> effect_B3
[1] -3.86
```

The average yield is about 90% at $B = 5$ and about 86% at $B = 7.5$, so the difference in mean yield between these two levels is approximately -3.9 percentage points: increasing B from 5 to 7.5 slightly decreases the yield.

One-factor model for B

To summarize this comparison, I fitted a simple first-order model in the coded variable x_B (R output from `summary(fit3)` and `anova(fit3)`):

```
> summary(fit3)

Call:
lm(formula = y3 ~ xB3, data = dat3)
```

```
Residuals:
     1      2      3      4 
 1.655 -0.585 -1.655  0.585 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  88.0250     0.8777 100.293 9.94e-05 ***
xB3          -1.9300     0.8777  -2.199    0.159    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.755 on 2 degrees of freedom
Multiple R-squared:  0.7074, Adjusted R-squared:  0.5611 
F-statistic: 4.836 on 1 and 2 DF,  p-value: 0.1589

> anova(fit3)
Analysis of Variance Table

Response: y3
          Df  Sum Sq Mean Sq F value Pr(>F)
xB3        1 14.8996 14.8996  4.8356 0.1589
Residuals  2  6.1625  3.0812               
```

The fitted regression in coded units is

$$\hat{y} = 88.03 - 1.93\, x_B,$$

so within this small range the model again suggests that lower B gives slightly higher yield. Because there are only four runs, the formal t-test for the slope is not significant at the 5% level, but the direction of the effect is consistent with the effect estimate and with the earlier submissions.

Residual plots

Figure 5 shows the residuals versus fitted values and the normal Q–Q plot for the one-factor model.

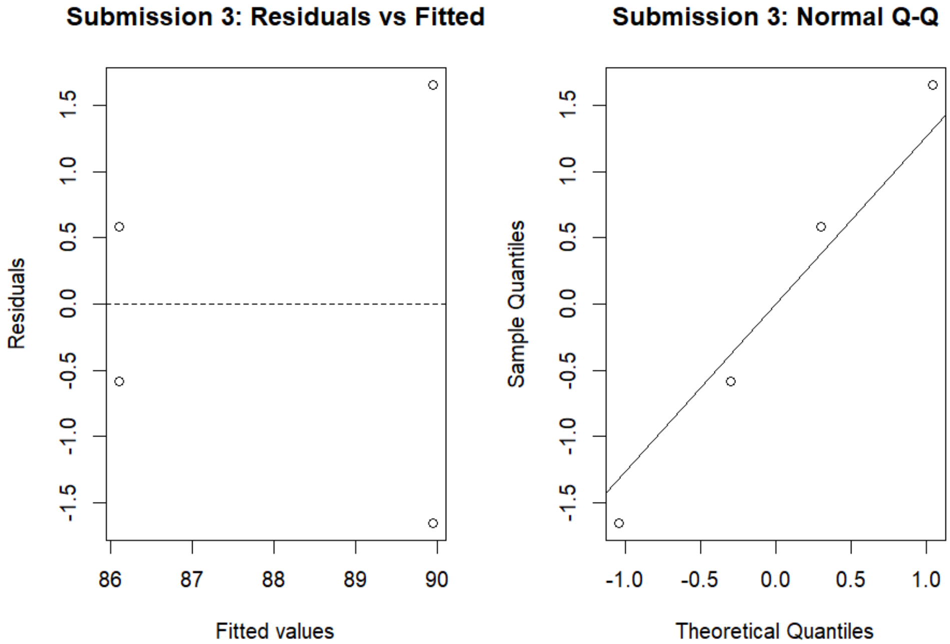

Figure 5: Submission 3: residuals vs fitted and normal Q–Q plot (R output from code #S3.4).

The residuals are small and show no obvious pattern, and the Q–Q plot is reasonably close to a straight line. This suggests that the simple linear model in x_B is adequate for summarizing the differences between $B = 5$ and $B = 7.5$ in this region.

Summary of Submission 3. Submission 3 confirms that the combination

$$B = 5, \quad D = 0, \quad F = 0, \quad A = C = E = G = H = 5$$

is a very high-yield setting, with all observed yields between about 85% and 92%. The small negative effect of B in this local comparison suggests that moving slightly below $B = 7.5$ is helpful, and that the high-yield region lies somewhere in the range between about $B = 5$ and $B = 7.5$, rather than at larger values of B.

Because D and F have already been pushed to their lower bounds and appear strongly detrimental when increased, I decided to keep $D = 0$ and $F = 0$ fixed for the next stage. The next submission (Submission 4) therefore focuses on exploring the effects of E and G around this high-yield region, using a 2^3 factorial in (B, E, G) plus center runs with the other factors held at $A = C = H = 5$ and $D = F = 0$.

3.4 Submission 4: 2^3 in (B, E, G) plus center runs

Submission 3 showed that $B = 5$, $D = 0$, and $F = 0$ with $A = C = E = G = H = 5$ is already a very good setting, with yields around 90%. In Submission 4 I kept

$$A = C = H = 5, \qquad D = 0, \qquad F = 0,$$

and explored the nearby cube in (B, E, G) using

$$B \in \{4, 6\}, \quad E \in \{2.5, 7.5\}, \quad G \in \{2.5, 7.5\},$$

together with four center runs at $B = E = G = 5$. This gives a 2^3 factorial in (B, E, G) plus center points, which is the first part of a face–centered CCD in these three factors.

Design and response

Table 4 shows the design matrix and the observed yield `ali4` (R output #S4.1).

Table 4: Submission 4: design matrix and response (raw R output)

```
> cbind(A4,B4,C4,D4,E4,F4,G4,H4,y4)
      A4 B4 C4 D4  E4 F4  G4 H4    y4
 [1,]  5  4  5  0 2.5  0 2.5  5 92.18
 [2,]  5  4  5  0 2.5  0 7.5  5 90.81
 [3,]  5  6  5  0 2.5  0 2.5  5 86.88
 [4,]  5  6  5  0 2.5  0 7.5  5 89.84
 [5,]  5  4  5  0 7.5  0 2.5  5 88.96
 [6,]  5  4  5  0 7.5  0 7.5  5 90.70
 [7,]  5  6  5  0 7.5  0 2.5  5 87.62
 [8,]  5  6  5  0 7.5  0 7.5  5 89.35
 [9,]  5  5  5  0 5.0  0 5.0  5 89.01
[10,]  5  5  5  0 5.0  0 5.0  5 91.33
[11,]  5  5  5  0 5.0  0 5.0  5 89.94
[12,]  5  5  5  0 5.0  0 5.0  5 89.87
```

The four center runs at $(B, E, G) = (5, 5, 5)$ are

$$y_{\text{center}} = 89.01,\ 91.33,\ 89.94,\ 89.87.$$

The overall range of the 12 responses is (R output #S4.4)

```
> range(y4)
[1] 86.88 92.18
```

so all runs in this local cube give yields between about 87% and 92%.

Factor-wise averages

The simple averages over the three levels of each factor (low, center, high) are (R output #S4.4):

```
> tapply(y4, B4, mean)
      4       5       6
90.6625 90.0375 88.4225

> tapply(y4, E4, mean)
    2.5       5     7.5
89.9275 90.0375 89.1575

> tapply(y4, G4, mean)
    2.5       5     7.5
```

```
88.9100 90.0375 90.1750

> mean(y4_center)
[1] 90.0375
```

For B the average yield is highest at the low level ($B = 4$) and lowest at the high level ($B = 6$), with the center ($B = 5$) in between. For E the differences are small, with a slight drop at $E = 7.5$. For G there is a mild increase as G moves from 2.5 to 7.5, but the effect is much smaller than the effect of B.

The mean of the four center runs is exactly 90.0375, which matches the value shown for the center level in the tables above.

First-order model in coded variables

To summarize the main trends in this cube I used coded variables x_B, x_E, x_G with values $-1, 0, +1$ at the low, center, and high levels of each factor. The first-order model (R output from `summary(fit4_fo)` and `anova(fit4_fo)`) is:

```
> summary(fit4_fo)

Call:
lm(formula = y4 ~ xB4_full + xE4_full + xG4_full, data = dat4)

Residuals:
    Min      1Q  Median      3Q     Max
-1.4600 -0.7356  0.1062  0.3050  1.6225

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  89.7075     0.3251 275.909   <2e-16 ***
xB4_full     -1.1200     0.3982  -2.813   0.0228 *
xE4_full     -0.3850     0.3982  -0.967   0.3619
xG4_full      0.6325     0.3982   1.588   0.1509
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.126 on 8 degrees of freedom
Multiple R-squared:  0.587, Adjusted R-squared:  0.4321
F-statistic: 3.789 on 3 and 8 DF,  p-value: 0.05854

> anova(fit4_fo)
Analysis of Variance Table

Response: y4
          Df  Sum Sq Mean Sq F value  Pr(>F)
xB4_full   1 10.0352 10.0352  7.9108 0.02275 *
xE4_full   1  1.1858  1.1858  0.9348 0.36194
xG4_full   1  3.2004  3.2004  2.5229 0.15086
Residuals  8 10.1484  1.2685
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The fitted regression function in coded variables is

$$\hat{y} = 89.71 - 1.12\, x_B - 0.39\, x_E + 0.63\, x_G.$$

In this small cube, the slope in B is negative and statistically significant at the 5% level, confirming that lower B gives slightly higher yield. The slopes in E and G are much smaller and not significant at the 5% level, though G shows a mild positive trend.

Center point check and curvature

To check how well the first-order model matches the center point, I computed the predicted mean at $x_B = x_E = x_G = 0$ and compared it to the average of the four center runs (R output #S4.7):

```
> pred_center
$fit
      1
89.7075

$se.fit
[1] 0.3251342

$df
[1] 8

$residual.scale
[1] 1.126298


> mean_center
[1] 90.0375
> diff_center
[1] 0.33
```

The model predicts 89.71 at the center, while the observed center mean is 90.04. The difference is only about 0.33 percentage points, which is small compared to the residual standard error (about 1.13). This suggests that there is no strong curvature exactly at the center, at least within this 2^3 cube.

Residual plots

Figure 6 shows the residuals versus fitted values and the normal Q–Q plot for the first-order model in (x_B, x_E, x_G).

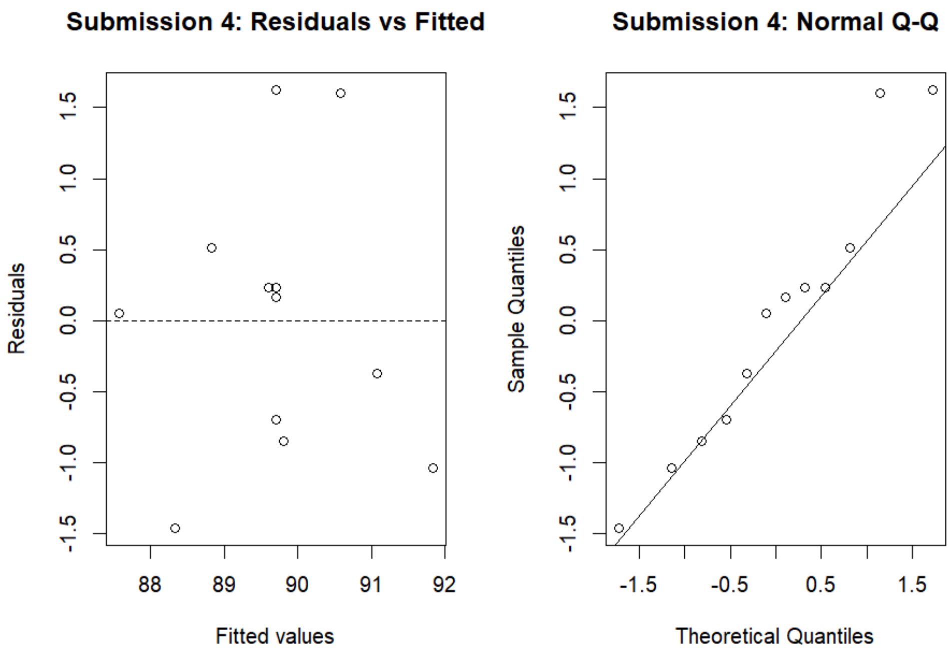

Figure 6: Submission 4: residuals vs fitted and normal Q–Q plot (R output from code #S4.6).

The residuals are small and show no clear pattern, and the Q–Q plot is close to a straight line. This supports the use of a simple linear model as a local approximation in this cube, even though the R^2 is lower than in Submission 2 because the overall variation in y is now much smaller.

Summary of Submission 4. Submission 4 confirms that the whole cube around $(B, E, G) = (5, 5, 5)$ with $A = C = H = 5$ and $D = F = 0$ gives very high yields, mostly between about 88% and 92%. Within this region:

- B still has a noticeable effect: lowering B from 6 down toward 4 increases the yield by roughly 1–2 percentage points.
- E has a very small effect in this neighborhood.
- G may have a mild positive effect, with slightly higher yields at larger G, but the effect is weak.

The best run so far is

$$B = 4, \quad E = 2.5, \quad G = 2.5, \quad A = C = H = 5, \quad D = F = 0,$$

with yield 92.18% (R output #S4.8). Submission 4 also provides four center runs and the eight factorial points needed for a face–centered CCD in (B, E, G). These results motivated Submission 5, which adds the six axial points needed so that a full second–order response-surface model in (B, E, G) can be fitted in this high-yield region.

3.5 Submission 5: axial points and quadratic model in (B, E, G)

Submission 5 keeps the same background levels

$$A = C = H = 5, \qquad D = F = 0,$$

and adds six axial points around the center $(B, E, G) = (5, 5, 5)$ to complete a small face–centered CCD in the three active factors.

Design and response

Table 5 shows the six axial runs in natural units.

Table 5: Submission 5: axial points in (B, E, G) (raw R output)

```
> dat5
  A5 B5 C5 D5  E5 F5  G5 H5    y5
1  5  4  5  0 5.0  0 5.0  5 89.55
2  5  6  5  0 5.0  0 5.0  5 90.51
3  5  5  5  0 2.5  0 5.0  5 92.17
4  5  5  5  0 7.5  0 5.0  5 92.67
5  5  5  5  0 5.0  0 2.5  5 89.86
6  5  5  5  0 5.0  0 7.5  5 91.03
```

Using coded variables $x_{B5}, x_{E5}, x_{G5} \in \{-1, 0, +1\}$ for low, center, and high levels, the same runs can be written as:

```
> dat5
     y5 xB5 xE5 xG5 B5  E5  G5
1 89.55  -1   0   0  4 5.0 5.0
2 90.51   1   0   0  6 5.0 5.0
3 92.17   0  -1   0  5 2.5 5.0
4 92.67   0   1   0  5 7.5 5.0
5 89.86   0   0  -1  5 5.0 2.5
6 91.03   0   0   1  5 5.0 7.5
```

The yields for these six axial runs range from about 89.6% to 92.7%. Some simple summaries are:

```
> tapply(y5, B5, mean)
      4       5       6
89.5500 91.4325 90.5100

> tapply(y5, E5, mean)
    2.5       5     7.5
92.1700 90.2375 92.6700

> tapply(y5, G5, mean)
   2.5      5    7.5
```

```
89.860 91.225 91.030

> range(y5)
[1] 89.55 92.67
> mean(y5)
[1] 90.965
```

At the end of Submission 5 the single best observed run is at $B = 5$, $E = 7.5$, $G = 5$, with yield 92.67%.

Axis means and comparison with the center

To compare the axial lines with the center point from Submission 4, I computed the mean at the center of the cube and the mean along each single factor axis (holding the other two at the center). The R output is:

```
> mean_center
[1] 90.0375

> cbind(B_axis_levels, y_B_axis)
     B_axis_levels y_B_axis
[1,]             4    89.55
[2,]             6    90.51

> cbind(E_axis_levels, y_E_axis)
     E_axis_levels y_E_axis
[1,]           2.5    92.17
[2,]           7.5    92.67

> cbind(G_axis_levels, y_G_axis)
     G_axis_levels y_G_axis
[1,]           2.5    89.86
[2,]           7.5    91.03

> mean_center
[1] 90.0375
> mean_B_axis
[1] 90.03
> mean_E_axis
[1] 92.42
> mean_G_axis
[1] 90.445
```

The mean along the B axis (89.55 at $B = 4$ and 90.51 at $B = 6$) is essentially the same as the center mean, which again suggests only a mild slope in B. The mean along the G axis is also very close to the center mean. Along the E axis the mean is higher (about 92.4%), because both $E = 2.5$ and $E = 7.5$ give yields above 92%. Overall, these numbers show that the region around $(B, E, G) = (5, 5, 5)$ is very flat and high, with all mean yields between about 90% and 92%.

Full CCD data set

Combining the eight factorial points and four center runs from Submission 4 with the six axial runs from Submission 5 gives an 18–run face–centered CCD in (B, E, G). In coded variables $x_B, x_E, x_G \in \{-1, 0, +1\}$ the combined data set (R output #S5.5) is:

```
> dat_CCD
       y xB xE xG B   E   G
1  92.18 -1 -1 -1 4 2.5 2.5
2  90.81 -1 -1  1 4 2.5 7.5
3  86.88  1 -1 -1 6 2.5 2.5
4  89.84  1 -1  1 6 2.5 7.5
5  88.96 -1  1 -1 4 7.5 2.5
6  90.70 -1  1  1 4 7.5 7.5
7  87.62  1  1 -1 6 7.5 2.5
8  89.35  1  1  1 6 7.5 7.5
9  89.01  0  0  0 5 5.0 5.0
10 91.33  0  0  0 5 5.0 5.0
11 89.94  0  0  0 5 5.0 5.0
12 89.87  0  0  0 5 5.0 5.0
13 89.55 -1  0  0 4 5.0 5.0
14 90.51  1  0  0 6 5.0 5.0
15 92.17  0 -1  0 5 2.5 5.0
16 92.67  0  1  0 5 7.5 5.0
17 89.86  0  0 -1 5 5.0 2.5
18 91.03  0  0  1 5 5.0 7.5

> nrow(dat_CCD)
[1] 18
```

The responses in this CCD range from 86.88% to 92.67%, and most runs are above 89%.

First-order and quadratic models in (x_B, x_E, x_G)

A first-order model in the coded variables x_B, x_E, x_G (R output from `summary(fit_ccd_fo)` and `anova(fit_ccd_fo)`) is:

```
> summary(fit_ccd_fo)

Call:
lm(formula = y ~ xB + xE + xG, data = dat_CCD)

Residuals:
    Min      1Q  Median      3Q     Max
-2.0817 -0.9547 -0.2992  0.9766  2.8013

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  90.1267     0.3318 271.636   <2e-16 ***
```

```
xB           -0.8000     0.4451  -1.797   0.0939 .
xE           -0.2580     0.4451  -0.580   0.5714
xG            0.6230     0.4451   1.400   0.1834
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.408 on 14 degrees of freedom
Multiple R-squared:  0.2829, Adjusted R-squared:  0.1293
F-statistic: 1.841 on 3 and 14 DF,  p-value: 0.186

> anova(fit_ccd_fo)
Analysis of Variance Table

Response: y
          Df  Sum Sq Mean Sq F value  Pr(>F)
xB         1  6.4000  6.4000  3.2298 0.09391 .
xE         1  0.6656  0.6656  0.3359 0.57140
xG         1  3.8813  3.8813  1.9587 0.18342
Residuals 14 27.7417  1.9815
```

The intercept 90.13 is the estimated mean at the center of the CCD, and the slope in x_B is negative (about -0.8), which again says that larger B tends to reduce the yield in this local region. The slopes in x_E and x_G are small and not significant. The R^2 is only about 0.28, because the total variation in y inside this already high region is quite small.

To allow for possible curvature and interactions, I also fitted a full second–order (quadratic) model in the three coded variables (this is the same model used as the final response surface in Section 2):

```
> fit_ccd_quad <- lm(
+   y ~ xB + xE + xG +
+     I(xB^2) + I(xE^2) + I(xG^2) +
+     xB:xE + xB:xG + xE:xG,
+   data = dat_CCD
+ )
> summary(fit_ccd_quad)

Call:
lm(formula = y ~ xB + xE + xG + I(xB^2) + I(xE^2) + I(xG^2) +
    xB:xE + xB:xG + xE:xG, data = dat_CCD)

Residuals:
     Min       1Q   Median       3Q      Max
-1.65179 -0.70527  0.02311  0.64395  1.90429

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  90.6618     0.5196 174.489  1.3e-15 ***
xB           -0.8000     0.4177  -1.915   0.0918 .
```

```
xE          -0.2580     0.4177  -0.618   0.5539
xG           0.6230     0.4177   1.492   0.1741
I(xB^2)     -1.2561     0.8024  -1.565   0.1561
I(xE^2)      1.1339     0.8024   1.413   0.1953
I(xG^2)     -0.8411     0.8024  -1.048   0.3252
xB:xE        0.4475     0.4670   0.958   0.3660
xB:xG        0.5400     0.4670   1.156   0.2809
xE:xG        0.2350     0.4670   0.503   0.6283
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.321 on 8 degrees of freedom
Multiple R-squared:  0.6393, Adjusted R-squared:  0.2335
F-statistic: 1.575 on 9 and 8 DF,  p-value: 0.2666

> anova(fit_ccd_fo, fit_ccd_quad)
Analysis of Variance Table

Model 1: y ~ xB + xE + xG
Model 2: y ~ xB + xE + xG + I(xB^2) + I(xE^2) + I(xG^2) + xB:xE + xB:xG +
    xE:xG
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     14 27.742
2      8 13.955  6    13.786 1.3172 0.3495
```

The quadratic model has a higher R^2 (about 0.64) and slightly smaller residual standard error ($\hat{\sigma} \approx 1.32$), but the overall F-test for the extra quadratic and interaction terms has $p \approx 0.35$. With only 18 runs, there is not enough evidence to say that the quadratic terms are statistically required, but the model still gives a smooth surface that is useful for contour plots, for describing the shape of the high-yield region, and for computing the stationary point used in Section 2.

Contour plot for (B, E)

Using the fitted quadratic model, I drew a contour plot of the predicted yield as a function of B and E, with G held at its center value. The plot (figure file `Sub4-5 contour.png`) is shown in Figure 7.

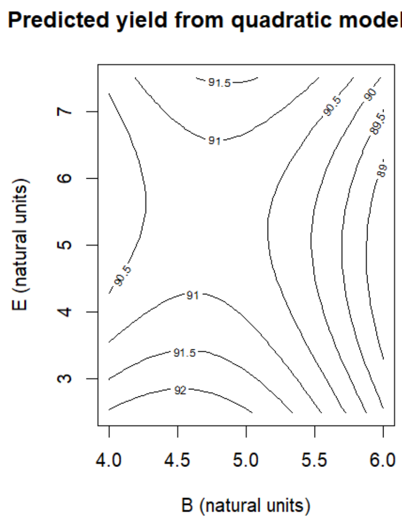

Figure 7: Submissions 4 and 5: contour plot of predicted yield from the quadratic model as a function of B and E (natural units) with G fixed at its center value.

The contour lines are almost vertical on the right-hand side of the plot, showing that larger B (moving toward 6) lowers the predicted yield. The contours are fairly flat in E, with a broad band of high predicted yield above about 91% when B is near 4–5 and E is between roughly 2.5 and 7.5. In other words, the quadratic model suggests a wide, flat high–yield region rather than a sharp peak.

Summary of Submission 5. Submission 5 completes a small face–centered CCD in (B, E, G) around the high–yield region found in earlier submissions. The six axial points show that:

- all runs in the CCD give yields between about 87% and 93%;
- at the end of Submission 5, the best observed run is at $B = 5$, $E = 7.5$, $G = 5$ with yield 92.67%;
- the response is quite flat near the center, with only a mild negative effect of B and very small effects of E and G in this region.

The quadratic model does not show strong, statistically significant curvature, but it provides a reasonable smooth surface and contour picture of the local high–yield region. This model is used in the final quadratic–model analysis in Section 3.7 to locate a stationary point inside the cube, and it motivated choosing $B = 5$, $E = 7.5$, $G = 5$ as a high–yield operating point for the replicate confirmation runs in Submission 6.

3.6 Submission 6: replicate confirmation runs at the chosen setting

After fitting the quadratic response surface in (B, E, G) and examining the contour and 3D plots, I selected the setting

$$A = C = H = 5, \quad D = F = 0, \quad B = 5, \quad E = 7.5, \quad G = 5$$

as a practical high–yield operating point. Submission 6 uses the remaining budget to run 12 replicates at this single setting in order to check stability and to obtain a prediction interval for future runs.

Design and response

All 12 runs use the same factor levels, and only the yield varies. The design matrix and responses (R object `ali6`) are:

```
> ali6

[1] 89.26 92.43 91.99 92.04 90.41 88.93 88.66 91.80 90.22 91.17 93.26 90.39

> cbind(A,B,C,D,E,F,G,H,ali6)

     A B C D   E F G H ali6
 [1,] 5 5 5 0 7.5 0 5 5 89.26
 [2,] 5 5 5 0 7.5 0 5 5 92.43
 [3,] 5 5 5 0 7.5 0 5 5 91.99
 [4,] 5 5 5 0 7.5 0 5 5 92.04
 [5,] 5 5 5 0 7.5 0 5 5 90.41
 [6,] 5 5 5 0 7.5 0 5 5 88.93
 [7,] 5 5 5 0 7.5 0 5 5 88.66
 [8,] 5 5 5 0 7.5 0 5 5 91.80
 [9,] 5 5 5 0 7.5 0 5 5 90.22
[10,] 5 5 5 0 7.5 0 5 5 91.17
[11,] 5 5 5 0 7.5 0 5 5 93.26
[12,] 5 5 5 0 7.5 0 5 5 90.39

> aliused
[1] 58
```

At the end of Submission 6, I have used 58 out of the 75 available runs.

Summary statistics and intervals

For these 12 replicates I computed the basic summary statistics and then a 95% confidence interval for the mean yield and a 95% prediction interval for a single future run at this setting. The R output (#S6.1 and #S6.3) is:

```
> length(y6)
[1] 12
> range(y6)
[1] 88.66 93.26
> mean(y6)
[1] 90.88
> sd(y6)
[1] 1.471765
> var(y6)
```

```
[1] 2.166091

> mean6; sd6; MSE6
[1] 90.88
[1] 1.471765
[1] 2.166091

> CI_mean6
[1] 89.94489 91.81511

> PI_one6
[1] 87.5084 94.2516
```

So the sample mean and standard deviation at the chosen setting are

$$\bar{y}_6 = 90.88, \qquad s_6 \approx 1.47.$$

The 95% confidence interval for the mean yield at this operating point is

$$(89.94,\ 91.82),$$

and the 95% prediction interval for a single future run at the same setting is

$$(87.51,\ 94.25).$$

These results, and in particular the 95% prediction interval $(87.51, 94.25)$, are the basis for the final performance statement at the recommended settings in Section 2.

Comparison with earlier center runs

To check whether the process behaves similarly to the earlier center runs in the CCD, I compared the mean and standard deviation from Submission 6 with the four center runs from Submission 4 at $(B, E, G) = (5, 5, 5)$. The R output (#S6.4) is:

```
> c(mean_center4 = mean_center4, mean_sub6 = mean6)
mean_center4    mean_sub6
     90.0375      90.8800

> c(sd_center4   = sd_center4,   sd_sub6  = sd6)
 sd_center4    sd_sub6
 0.9598394  1.4717646
```

The mean yield from Submission 6 (about 90.9%) is slightly higher than the mean of the earlier center runs (about 90.0%), which is consistent with moving to a slightly better point in the flat high–yield region. The standard deviation is also a bit larger, but both sets of runs show variability on the order of 1–1.5 percentage points, which is small compared to the overall level of the yield.

Run-order and normality checks

I also checked the run–order plot and the normal Q–Q plot for the 12 replicates at this setting. The histogram and Q–Q plot (figure file `Sub6- Histogram and NormalQQ.png`) are shown in Figure 8, and the run–order plot (figure file `Sub6- RunOrder.png`) is shown in Figure 9.

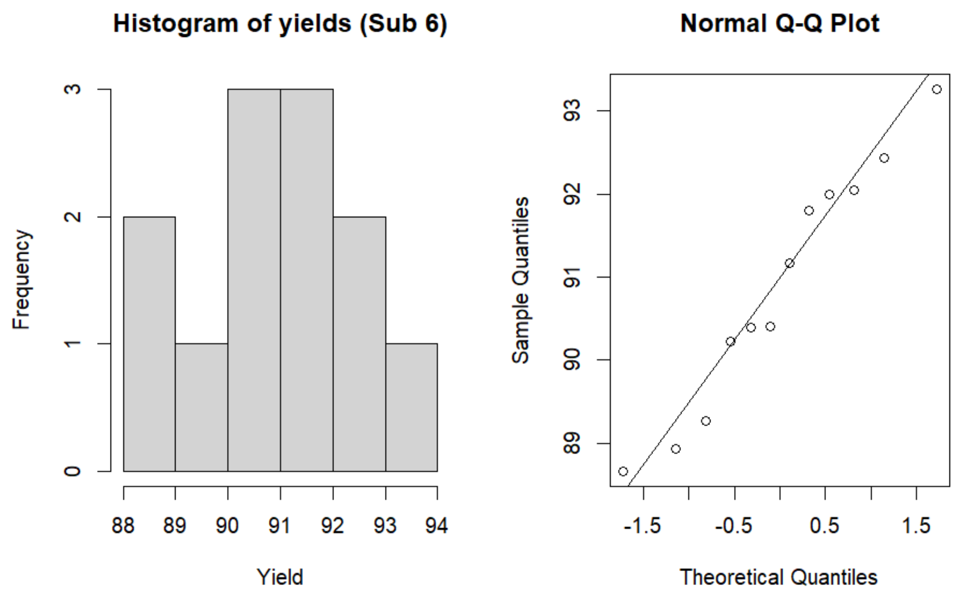

Figure 8: Submission 6: histogram of the 12 replicate yields and normal Q–Q plot for the replicate residuals.

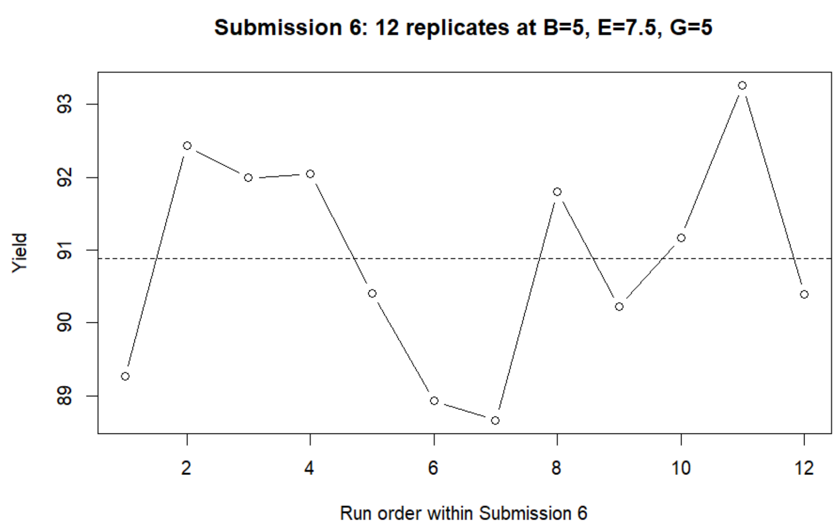

Figure 9: Submission 6: yield versus run order at the chosen operating point.

The histogram and Q–Q plot show no clear departures from approximate normality, and the run–order plot does not show any obvious drift or trend over time. This supports the assumption of a stable process with roughly constant variance at the chosen operating point.

Summary of Submission 6. Submission 6 confirms that the recommended setting

$$A = C = H = 5, \quad D = F = 0, \quad B = 5, \quad E = 7.5, \quad G = 5$$

gives a stable, high yield with mean about 91% and typical run-to-run variation of about 1–2 percentage points. The 95% prediction interval $(87.51, 94.25)$ from these replicates is used in Section 2 as the final statement of expected performance at the chosen optimum.

3.7 Final quadratic model analysis in (B, E, G)

In the final step I used all 18 runs from the (B, E, G) CCD (Submissions 4 and 5) to fit a full second–order response–surface model in the coded variables

$$x_B, x_E, x_G \in \{-1, 0, +1\},$$

with the other factors fixed at

$$A = C = H = 5, \qquad D = F = 0.$$

This is the model used in Section 2 to summarize the local behavior of the process and to compare candidate operating points.

Quadratic model in coded variables

The final quadratic model (R object `quad_fit`) has the form

$$\hat{y} = \beta_0 + \beta_B x_B + \beta_E x_E + \beta_G x_G + \beta_{BB} x_B^2 + \beta_{EE} x_E^2 + \beta_{GG} x_G^2 + \beta_{BE} x_B x_E + \beta_{BG} x_B x_G + \beta_{EG} x_E x_G,$$

where y is the yield in percent. The R output (#FinalRuns1.1) is:

```
> summary(quad_fit)

Call:
lm(formula = y ~ xB + xE + xG + I(xB^2) + I(xE^2) + I(xG^2) +
    xB:xE + xB:xG + xE:xG, data = dat_CCD)

Residuals:
     Min       1Q   Median       3Q      Max
-1.65179 -0.70527  0.02311  0.64395  1.90429

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  90.6618     0.5196 174.489  1.3e-15 ***
xB           -0.8000     0.4177  -1.915   0.0918 .
xE           -0.2580     0.4177  -0.618   0.5539
xG            0.6230     0.4177   1.492   0.1741
I(xB^2)      -1.2561     0.8024  -1.565   0.1561
```

```
I(xE^2)       1.1339     0.8024   1.413   0.1953
I(xG^2)      -0.8411     0.8024  -1.048   0.3252
xB:xE         0.4475     0.4670   0.958   0.3660
xB:xG         0.5400     0.4670   1.156   0.2809
xE:xG         0.2350     0.4670   0.503   0.6283
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.321 on 8 degrees of freedom
Multiple R-squared:  0.6393, Adjusted R-squared:  0.2335
F-statistic: 1.575 on 9 and 8 DF,  p-value: 0.2666

> anova(quad_fit)
Analysis of Variance Table

Response: y
          Df  Sum Sq Mean Sq F value  Pr(>F)
xB         1  6.4000  6.4000  3.6689 0.09176 .
xE         1  0.6656  0.6656  0.3816 0.55393
xG         1  3.8813  3.8813  2.2250 0.17413
I(xB^2)    1  5.3290  5.3290  3.0549 0.11862
I(xE^2)    1  2.1639  2.1639  1.2405 0.29772
I(xG^2)    1  1.9168  1.9168  1.0988 0.32515
xB:xE      1  1.6021  1.6021  0.9184 0.36596
xB:xG      1  2.3328  2.3328  1.3373 0.28087
xE:xG      1  0.4418  0.4418  0.2533 0.62835
Residuals  8 13.9553  1.7444
```

No single term is strongly significant at the 5% level, which matches the visual impression that the whole (B, E, G) region is fairly flat and high. However, the model still gives a smooth surface that is useful for contour plots and for comparing candidate settings.

Stationary point and nature of the surface

Using the quadratic part of the model, I computed the stationary point in the coded variables and the eigenvalues of the curvature matrix. The R output (#FinalRuns2.2) is:

```
> x_star
       xB*        xE*        xG*
-0.2283254  0.1262082  0.3146957

> eig_B
[1]  2.329757 -1.460083 -2.796103

> c(B_star = B_star, E_star = E_star, G_star = G_star)
B_star.xB* E_star.xE* G_star.xG*
  4.771675   5.315520   5.786739

> pred_star$fit
```

```
     xB*
90.83486
> pred_star$se.fit
[1] 0.5204182
```

The stationary point in coded units is approximately

$$(x_B^*, x_E^*, x_G^*) \approx (-0.23,\ 0.13,\ 0.31),$$

which corresponds in natural units to

$$B^* \approx 4.77, \quad E^* \approx 5.32, \quad G^* \approx 5.79.$$

The eigenvalues (2.33, −1.46, −2.80) are mixed in sign, so the stationary point is a saddle rather than a true local maximum or minimum. The predicted mean at this point is about 90.8%, which is actually a little lower than the best observed runs in the CCD.

This supports the idea that the response surface is very flat and high in a broad region around the middle of the cube, with no sharp interior peak in (B, E, G).

In the RSM notes, extra runs are often used for replicates at the stationary point of the quadratic model. The calculated stationary point is $(B^*, E^*, G^*) \approx (4.77, 5.32, 5.79)$ with predicted yield 90.8%. However, the highest observed yield in the CCD (92.67%) occurred at $(B, E, G) = (5, 7.5, 5)$. Because the stationary point is a saddle with only a slightly lower predicted mean, and the region is quite flat, I chose to replicate at the empirically best setting instead of at the saddle.

Comparison of candidate settings

To compare the center of the design, the best observed CCD point from Submission 4, and the chosen operating point, I evaluated the quadratic model at three coded settings. The R output (#FinalRuns2.3) is:

```
> new_pts
                            xB xE xG
center: B=5,E=5,G=5          0  0  0
Sub4 best: B=4,E=2.5,G=2.5  -1 -1 -1
chosen point: B=5,E=7.5,G=5  0  1  0

> data.frame(
+   setting   = rownames(new_pts),
+   xB        = new_pts$xB,
+   xE        = new_pts$xE,
+   xG        = new_pts$xG,
+   fit       = as.numeric(pred_pts$fit),
+   se_fit    = as.numeric(pred_pts$se.fit)
+ )
                      setting xB xE xG      fit    se_fit
1         center: B=5,E=5,G=5  0  0  0 90.66179 0.5195848
2  Sub4 best: B=4,E=2.5,G=2.5 -1 -1 -1 91.35607 1.1769220
3 chosen point: B=5,E=7.5,G=5  0  1  0 91.53771 0.9383561
```

The quadratic model predicts:

- about 90.7% at the CCD center $(B, E, G) = (5, 5, 5)$;
- about 91.4% at the best observed CCD point from Submission 4, $(4, 2.5, 2.5)$;
- about 91.5% at the chosen point $(5, 7.5, 5)$.

The differences are only about 1 percentage point, which is small relative to the residual standard error of the model ($\hat{\sigma} \approx 1.32$). This confirms that all three settings lie in a broad, flat high–yield region, and that choosing $(B, E, G) = (5, 7.5, 5)$ is reasonable and consistent with both the model and the observed data.

Residual diagnostics for the final model

Finally, I checked the residuals from the quadratic model `quad_fit`. Figure 10 shows the residuals versus fitted values and the normal Q–Q plot (figure file `FinalResidual.png`).

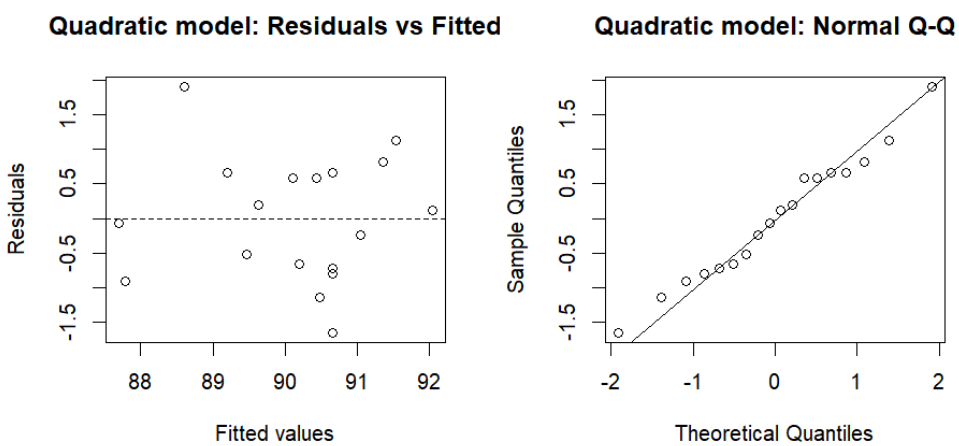

Figure 10: Residuals vs fitted values and normal Q–Q plot for the final quadratic model in (x_B, x_E, x_G).

The residuals are small (mostly within about ± 2 percentage points), show no clear pattern against the fitted values, and the Q–Q plot is close to a straight line. This supports the usual linear–model assumptions and suggests that the quadratic response surface is an adequate summary of the behavior of the process in the high–yield region.

Overall, this final model analysis agrees with the main conclusions:

- the process has a wide, flat high–yield region in (B, E, G);
- larger B tends to reduce yield slightly, while E and G have only mild effects in this local neighborhood;
- the chosen operating point $A = C = H = 5$, $D = F = 0$, $B = 5$, $E = 7.5$, $G = 5$ is inside this flat, high region and is supported by both the model and the replicate runs from Submission 6.

4 Learning experience

This project was a very good learning experience for me. At the beginning I spent quite a lot of time just trying to fully understand the problem and to shape a reasonable plan for my runs. Each time I sent a submission and received the results back I was honestly excited to see what happened, and then I took time to carefully analyze the output before deciding on the next step. Sometimes this was a little stressful and annoying, because I did not want to send a "bad" or wasteful submission, so I kept checking that each new design made sense and stayed within the budget.

Working through the stages of screening, follow–up designs, and the small response–surface experiment helped me understand the ideas from class much better. Concepts like active factors, curvature, CCDs, and prediction intervals became much clearer once I had to use them to make real decisions about where to run next. I am also happy that I was able to find a stable high–yield region for the process and build a sensible final recommendation. Finally, I learned how important it is to organize my R work: throughout the project I labeled each piece of output and kept the code in a clean structure so that I could later plug the results into this report in a neat and consistent way.